

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 7 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTEEOE705A - TESTING, MAINTENANCE AND COMMISSIONING OF ELECTRICAL EQUIPMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
-

Q1. Objective type questions. (Compulsory Question) 12

- 1 While conducting short-circuit test on a transformer which side is short circuited?
(A) High voltage side (B) Low voltage side
(C) Primary side (D) Secondary side
- 2 Short circuit test on transformers is conducted to determine _____
(A) Core losses (B) Copper losses
(C) Hysteresis losses (D) Eddy current losses
- 3 During open circuit test (OC) of a transformer _____
(A) primary is supplied rated kVA
(B) primary is supplied full-load current
(C) primary is supplied full-load current
(D) primary is supplied rated voltage
- 4 Earth electrode provides _____
(A) high resistance (B) medium resistance
(C) low resistance (D) very high resistance
- 5 Which of the following parameters don't define that transformer is built correctly?
(A) Losses (B) Polarity (C) Tap changing operation
(D) Ratio
- 6 Which of the following is the reason relating to the maintenance while operation?
(A) To obtain the maximum practicable operating efficiency
(B) To obtain optimum life
(C) To minimize the risk of premature and unexpected failure
(D) Maximum efficiency, life, minimum temperature
- 7 An induction motor can be said analogous to
(A) transformer (B) synchronous motor (C) universal motor
(D) stepper motor

- 8 When the rotor of 3 phase induction motor run at synchronous speed, its rotor frequency will become
 (A) Zero (B) Half of supply frequency. (C) Equal to supply frequency.
 (D) Very high.
- 9 Grading of cables
 (A) Reduces insulation cost and increases current rating
 (B) Reduces insulation cost but decreases current rating
 (C) Increases both
 (D) Either A or B
- 10 Specific resistance of soil is _____
 (A) Changes from soil to soil
 (B) is constant
 (C) Depends on the circuit connected to it
 (D) Depends on the supply voltage
- 11 Protective relays can be designed to respond to _____
 (A) Light intensity, impedance (B) Temperature, resistance, reactance
 (C) Voltage and current (D) All of these
- 12 Which component ensures the safety of the line from damage?
 (A) Relay (B) Circuit breaker (C) Bus bar (D) Current transformer

Q2. Solve the following.

- A) Explain the importance and necessity of the maintenance. 6
- B) What are the Different Types of Maintenance? Write Short notes on preventive Maintenance and breakdown Maintenance. 6

Q3. Solve the following.

- A) Explain Condition monitoring of transformer bushings. 6
- B) Explain dissolved gas analysis for transformer oil. 6

Q4. Solve Any Two of the following.

- A) Explain induction motor fault monitoring method and Remedies. 6
- B) Explain Type tests, Routine tests of transformer. 6
- C) Write the recommended maintenance schedule for induction motor as per IS-900 6

Q5. Solve Any Two of the following.

- A) Explain the various test conducted on Power Cable at work and its site. 6
- B) Explain Testing of capacitor banks 6
- C) Describe the various causes of failure of power cable. 6

Q6. Solve Any Two of the following.

- A) What are different electrode configuration for substation earthing system. explain any one method of earthing. 6
- B) Explain in details ultrasonic tests to detect internal mechanical faults. 6
- C) Explain the touch voltage, step voltage, Tolerable step and Touch Voltages. 6

*** End ***